dirac16complex: symbols and their definitions

A dictionary of every symbol used for the Dirac equation and the Kohn–Sham theories of the 16-component quantized fermion field dirac16complex on 7+1 spacetime — sections 7–10 of the notebook 7+1.nb

Repository: github.com/once-ere/7plus1 — prepared for Nash
Notebook: 7+1.nb (Wolfram Language 15.0.1)
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Abstract

For each symbol this dictionary gives the mathematical notation, the name used in the Mathematica notebook 7+1.nb (where there is one), the defining equation, and its meaning. The symbols are grouped by topic in the order in which they are introduced; longer defining equations follow each table as displayed formulas. Section 10 lists symbols that look alike, and section 11 lists every Mathematica name defined in sections 7–10 of 7+1.nb with its Wolfram Language definition. Every equation needed to define a symbol is written in this document.

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1. Conventions

- Units: $\hbar = c = 1$.
- Index ranges: $A, B, K, L \in \{0, 1, \ldots, 7\}$ (spacetime), $i, j \in \{1, \ldots, 7\}$ (space), $a, b, c, d \in \{1, \ldots, 16\}$ (spinor components), $s, r \in \{1, \ldots, 8\}$ (spin labels), k, l (orbitals or Grassmann generators). Repeated indices are summed unless stated otherwise.
- Indices are raised and lowered with the metric η of section 2: $a_A = \eta_{AA} a^A$ (no sum), so $a_0 = a^0$ and $a_i = -a^i$.
- Matrices: 1_n is the $n \times n$ identity, M^{T} the transpose, $\overline{M}$ or M^* the complex conjugate, $M^\dagger = \overline{M}^{\mathsf{T}}$ the Hermitian conjugate, tr the trace, δ_{ij} the Kronecker delta, $\{X, Y\} = XY + YX$ the anticommutator and $[X, Y] = XY - YX$ the commutator.

- Block notation: $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ is the 16×16 matrix with the four 8×8 blocks a, b, c, d; a 16-component column is written $(\psi_\uparrow; \psi_\downarrow)$, upper 8 components first.
- Typefaces: Kohn–Sham potentials are calligraphic ($\mathcal{S}, \mathcal{P}, \mathcal{V}_A, \mathcal{A}_A$), external sources are sans-serif ($\mathsf{s}, \mathsf{p}, \mathsf{v}_A, \mathsf{a}_A$). Mathematica names are set in `typewriter`.
- Mathematica lists are 1-based: η_{AB} is `η17[[A+1, B+1]]`, ψ_a is `dirac16complex[[a]]`. Greek letters in names are Mathematica named characters: `\[Psi]` is ψ, `\[Phi]` is ϕ, `\[CapitalGamma]` is Γ, and so on.
- Wirtinger convention: a complex variable z and its conjugate z^* are treated as independent variables when differentiating, $\partial z/\partial z = 1$, $\partial z^*/\partial z = 0$.

2. Spacetime and momenta

symbol	Mathematica	definition	meaning
X; $x_0, \ldots, x_7$	`X = {x0, …, x7}`	$X = (x_0, x_1, \ldots, x_7)$, $x^A = x_A$ as symbols	x_0 ordinary time (direction of the octonion unit $e_0 = 1$); x_1, x_2, x_3 ordinary 3-space; $x_4, \ldots, x_7$ auxiliary space-like
x, $\vec{x}$	—	$x = (x_0, \vec{x})$, $\vec{x} = (x_1, \ldots, x_7)$	spacetime point; space part
η_{AB}, η^{AB}	`η17`, `η17inv`	$\eta =$ diag$(+1, -1, -1, -1, -1, -1, -1, -1)$, $\eta^{-1} = \eta$	metric of 7+1 spacetime: simply the squares of the octonion basis, $\eta_{AB} = \mathrm{Re}(e_A e_B)$, $e_0^2 = +1$, $e_i^2 = -1$
η'	`η71`	$\eta' = -\eta =$ diag$(-1, +1, \ldots, +1)$	the same spacetime in the sign convention of 4+4.nb (space-like +1); its relation has no real 16×16 realization
$\{7, 1, 0\}$	`DefMetric[{7, 1, 0}, g71[-a, -b], CD, …]`	xAct signature list (section 12 of 7+1.nb)	seven space-like dimensions, one time-like dimension, none degenerate
$a \cdot b$	—	$\eta_{AB} a^A b^B = a^0 b^0 - \sum_i a^i b^i$	Minkowski product
∂_A	`D[f, X[[A+1]]]`	$\partial_A = \partial/\partial x_A$, $\partial_A x^B = \delta_A^B$	partial derivative
$\Box$	`waveOp17`	$\eta^{AB}\partial_A\partial_B = \partial_0^2 - \partial_1^2 - \cdots - \partial_7^2$	wave (d'Alembert) operator
$\vec{p}$, p^i	`pv7`	$\vec{p} = (p^1, \ldots, p^7)$; `pv7 = {p1, …, p7}`	momentum; real symbols in the symbolic checks
$\vec{p} \cdot \vec{x}$, $\vec{p}^2$	—	$\sum_i p^i x_i$, $\sum_i (p^i)^2$	Euclidean products of space vectors
E	`Ep`	$E = \sqrt{m^2 + \vec{p}^2}$	energy of a free particle of momentum $\vec{p}$
p	—	$p = (E, \vec{p})$, $p \cdot p = m^2$	8-momentum on the mass shell
e_s	`UnitVector[8, s]`	$(e_s)_r = \delta_{rs}$	unit column of length 8
$\delta^7(\vec{x} - \vec{y})$	—	$\int \delta^7(\vec{x} - \vec{y}) f(\vec{y})\, d^7y = f(\vec{x})$	Dirac delta in 7 space dimensions

The Minkowski square of a point is the real part of the square of the octonion $x = x_0 e_0 + \sum_i x_i e_i$:

$$x \cdot x = \eta_{AB} x^A x^B = x_0^2 - x_1^2 - \cdots - x_7^2 = \mathrm{Re}(x\, x).$$

The signature: $\{7, 1, 0\}$ and diag$(+1, -1, \ldots, -1)$. The 7+1 spacetime has one time-like dimension (x_0, the direction of the octonion unit $e_0 = 1$) and seven space-like dimensions (x_1, x_2, x_3 ordinary space, $x_4, \ldots, x_7$ auxiliary space). Two different objects describe this, and neither may be replaced by the other. (i) The tensor package xAct, used in section 12 of 7+1.nb exactly as in the author's notebook Einstein-Lovelock-7+1-Nash.nb, declares the metric with `DefMetric[{7, 1, 0}, g71[-a, -b], CD, …]`: the list $\{7, 1, 0\}$ declares seven space-like dimensions, one time-like dimension and no degenerate dimension — the numbers of entries $+1$ and -1 of $\eta' =$ diag$(-1, +1, \ldots, +1)$ (`η71`), the same spacetime written in the sign convention of 4+4.nb

(space-like = +1, time-like = −1). (ii) The matrix $\eta = -\eta' = \mathrm{diag}(+1,-1,\ldots,-1)$ (`η17`) is simply the squares of the octonion basis: $e_0^2 = +1$ for the unit $e_0 = 1$ (time) and $e_i^2 = -1$ for the seven imaginary units $e_1,\ldots,e_7$ (space); more generally it is the real part of the product of two basis units, and it gives the quadratic form of the octonions:

$$\eta_{AB} = \mathrm{Re}(e_A e_B), \qquad e_0^2 = +1, \qquad e_i^2 = -1 \;\; (i = 1,\ldots,7),$$
$$\mathrm{Re}(x\,x) = x_0^2 - x_1^2 - \cdots - x_7^2 = \eta_{AB}\, x^A x^B \qquad (x = x^A e_A \in \mathbb{O}).$$

η gives the signs of the Clifford relation $\Gamma_A\Gamma_B + \Gamma_B\Gamma_A = 2\,\eta_{AB}\,1_{16}$ of the real 16×16 Dirac matrices: time squares to $+1_{16}$ and each of the seven space directions to -1_{16}, exactly like the octonion units (in the notation $Cl(p,q)$, with p generators squaring to +1 and q to −1, the algebra $Cl(1,7) \cong \mathbb{R}(16)$). It also gives the signs of the metric components: $g_{00} > 0$ along the time-like direction, negative squares along the space-like directions. Every component computed by 7+1.nb follows from the component matrix of the metric; the declared list fixes for xAct the sign of the determinant of the metric, `SignDetOfMetric[g71]` = −1, which is the same in both sign conventions, since $\det\eta = \det\eta' = -1$. 7+1.nb checks `SignatureOfMetric[g71]` = {7, 1, 0}, `SignDetOfMetric[g71]` = −1, $\mathrm{Re}(e_A e_B) = \eta_{AB}$ and $\mathrm{Re}(x\,x) = \eta_{AB}x^A x^B$.

3. The real 8×8 matrices

symbol	Mathematica	definition	meaning
τ_i^O	`τO[i]`	the seven real 8×8 matrices displayed below; $\tau_0^O = 1_8$	left multiplication by the imaginary octonion unit e_i, written in a spinor basis
τ_A	`τ[A]`	$\tau_0 = 1_8$; $\tau_i = \tau_i^O$ $(i \le 3)$; $\tau_4,\ldots,\tau_7$ displayed below	the real split-octonion matrices of the notebook 4+4.nb
σ	`σ`	$\sigma = \tau_1\tau_2\tau_3 = \left(\begin{smallmatrix} 0 & 1_4 \\ 1_4 & 0 \end{smallmatrix}\right)$	the 8×8 matrix with $\tau_k^O = \tau_k\sigma$ for $k = 4,\ldots,7$
$\bar\tau_A$	`τbar[A]`	$\bar\tau_0 = 1_8$, $\bar\tau_A = \sigma\,\tau_A^{\mathsf T}\sigma$ $(A = 1,\ldots,7)$	conjugate split-octonion matrices of 4+4.nb
$\eta^{(4,4)}$	`η4488`	$\mathrm{diag}(+1,+1,+1,+1,-1,-1,-1,-1)$	metric of the notebook 4+4.nb
T_A	`T16[A]`	$T_A = \left(\begin{smallmatrix} 0 & \bar\tau_A \\ \tau_A & 0 \end{smallmatrix}\right)$, $\{T_A, T_B\} = 2\eta_{AB}^{(4,4)} 1_{16}$	the 16×16 Dirac matrices of 4+4.nb
σ_i	`σW[i]`	$\sigma_i = i\,\tau_i^O$	Hermitian; the Pauli matrices of 7 space dimensions
$\tau^O\cdot\vec p$	—	$\sum_i p^i \tau_i^O$	real, antisymmetric
$\sigma\cdot\vec p$	`σWp[p]`	$\sum_i p^i \sigma_i$	Hermitian
σ^A, $\bar\sigma^A$	`σUp`, `σbarUp`	$\sigma^A = (1_8, \sigma_1, \ldots, \sigma_7)$, $\bar\sigma^A = (1_8, -\sigma_1, \ldots, -\sigma_7)$	chiral 8×8 matrices (lists of 8 matrices; `σUp[[A+1]]` is σ^A)

Defining relations $(i, j = 1,\ldots,7)$:

$$(\tau_i^O)^{\mathsf T} = -\tau_i^O, \qquad \tau_i^O\tau_j^O + \tau_j^O\tau_i^O = -2\,\delta_{ij}\,1_8, \qquad \tau_1^O\tau_2^O\cdots\tau_7^O = 1_8,$$
$$\sigma_i^\dagger = \sigma_i, \qquad \sigma_i\sigma_j + \sigma_j\sigma_i = 2\,\delta_{ij}\,1_8, \qquad (\tau^O\cdot\vec p)^2 = -\vec p^{\,2}\,1_8, \qquad (\sigma\cdot\vec p)^2 = \vec p^{\,2}\,1_8,$$
$$\sigma^2 = 1_8, \qquad \sigma\tau_k = \tau_k\sigma \;\; (k = 1,2,3), \qquad \sigma\tau_k = -\tau_k\sigma, \;\; \tau_k^2 = +1_8 \;\; (k = 4,\ldots,7).$$

The seven matrices τ_i^O:

$$\tau_1^O = \begin{pmatrix} 0&0&0&0&0&0&0&1\\ 0&0&0&0&0&0&1&0\\ 0&0&0&0&0&-1&0&0\\ 0&0&0&0&-1&0&0&0\\ 0&0&0&1&0&0&0&0\\ 0&0&1&0&0&0&0&0\\ 0&-1&0&0&0&0&0&0\\ -1&0&0&0&0&0&0&0 \end{pmatrix}, \quad \tau_2^O = \begin{pmatrix} 0&0&0&0&0&0&-1&0\\ 0&0&0&0&0&0&0&1\\ 0&0&0&0&1&0&0&0\\ 0&0&0&0&0&-1&0&0\\ 0&0&-1&0&0&0&0&0\\ 0&0&0&1&0&0&0&0\\ 1&0&0&0&0&0&0&0\\ 0&-1&0&0&0&0&0&0 \end{pmatrix},$$

$$\tau_3^O = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad \tau_4^O = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \end{pmatrix},$$

$$\tau_5^O = \begin{pmatrix} 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \end{pmatrix}, \quad \tau_6^O = \begin{pmatrix} 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \end{pmatrix},$$

$$\tau_7^O = \begin{pmatrix} 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

The split-octonion matrices $\tau_4, \ldots, \tau_7$ of 4+4.nb (they square to $+1_8$; $\tau_k^O = \tau_k\, \sigma$):

$$\tau_4 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad \tau_5 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix},$$

$$\tau_6 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad \tau_7 = \begin{pmatrix} -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

4. Dirac matrices, chirality and the chiral projections

symbol	**Mathematica**	**definition**	**meaning**
Γ_A	`Γ17[A]`	$\Gamma_0 = \left(\begin{smallmatrix} 0 & 1_8 \\ 1_8 & 0 \end{smallmatrix}\right)$, $\Gamma_i = \left(\begin{smallmatrix} \tau_i^O & 0 \\ 0 & -\tau_i^O \end{smallmatrix}\right)$	the eight REAL 16×16 Dirac matrices; every entry is 0 or ±1
Γ^A	`Γ17up[A]`	$\Gamma^A = \eta^{AA}\Gamma_A$: $\Gamma^0 = \Gamma_0$, $\Gamma^i = -\Gamma_i$	Dirac matrix with an upper index
Γ_K	`products256`	$\Gamma_K = \Gamma_{A_1}\Gamma_{A_2}\cdots\Gamma_{A_k}$ for $K = \{A_1 < \cdots < A_k\}$; $\Gamma_\emptyset = 1_{16}$	the 256 basis elements of $Cl(1,7) \cong \mathbb{R}(16)$
β	`βD`	$\beta = \Gamma^0 = \left(\begin{smallmatrix} 0 & 1_8 \\ 1_8 & 0 \end{smallmatrix}\right)$	Dirac β; real symmetric
α_i	`αD[i]`	$\alpha_i = \Gamma^0\Gamma^i = \left(\begin{smallmatrix} 0 & \tau_i^O \\ -\tau_i^O & 0 \end{smallmatrix}\right)$	Dirac α; real symmetric
ω	`ω17`, `Γ17[8]`	$\omega = \Gamma_0\Gamma_1\cdots\Gamma_7 = \left(\begin{smallmatrix} 0 & -1_8 \\ 1_8 & 0 \end{smallmatrix}\right)$	volume element; real, $\omega^2 = -1_{16}$
Σ_{AB}	`Σ17[A, B]`	$\Sigma_{AB} = \frac{1}{4}(\Gamma_A\Gamma_B - \Gamma_B\Gamma_A)$	generators of Spin(1,7); real
χ	`χ17`	$\chi = i\,\omega = \left(\begin{smallmatrix} 0 & -i1_8 \\ i1_8 & 0 \end{smallmatrix}\right)$	chirality matrix; Hermitian, complex
P_L, P_R	`PL17`, `PR17`	$P_L = \frac{1}{2}(1_{16} - \chi)$, $P_R = \frac{1}{2}(1_{16} + \chi)$	chiral projections; complex, rank 8 each
E_L, E_R	`EL`, `ER`	$E_L = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1_8 \\ -i1_8 \end{smallmatrix}\right)$, $E_R = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} -i1_8 \\ 1_8 \end{smallmatrix}\right)$	16×8 isometries onto the eigenspaces $\chi = -1$ and $\chi = +1$
W	`Wch`	$W = (E_L, E_R)$	unitary 16×16 change to the chiral basis
η'	—	$\eta' = -\eta = \mathrm{diag}(-1, +1, \ldots, +1)$	the opposite sign convention; generates $Cl(7,1) \cong \mathbb{H}(8)$, which has no real 16×16 realization
R_A, T, Y	—	hypothetical real R_A with $R_AR_B + R_BR_A = 2\eta'_{AB}1_{16}$; $R_A = T(i\Gamma_A)T^{-1}$; $Y = T^{-1}\overline{T}$	symbols of the proof that η' has no real 16×16 realization (Y would satisfy $Y\overline{Y} = 1_{16}$ and $Y = c\,\omega$, impossible since $\omega^2 = -1_{16}$)
Z	—	real 16×16 matrix with $Z\Gamma_A = -\Gamma_A Z$ $(A = 0, \ldots, 7)$	the solutions are $Z = c\,\omega$, $c \in \mathbb{R}$, and $Z^2 = -c^2 1_{16}$: there is no real chirality matrix

Defining relations ($A, B = 0, \ldots, 7$; $i, j = 1, \ldots, 7$):

$$\Gamma_A\Gamma_B + \Gamma_B\Gamma_A = 2\,\eta_{AB}\,1_{16}, \qquad \Gamma_0^2 = 1_{16}, \quad \Gamma_i^2 = -1_{16}, \qquad \Gamma^A\Gamma_A = 8\cdot 1_{16},$$
$$(\Gamma^A)^\dagger = (\Gamma^A)^\mathsf{T} = \Gamma^0\Gamma^A\Gamma^0,$$
$$\alpha_i\alpha_j + \alpha_j\alpha_i = 2\,\delta_{ij}\,1_{16}, \qquad \alpha_i\beta + \beta\alpha_i = 0, \qquad \beta^2 = 1_{16},$$
$$\omega\,\Gamma_A = -\Gamma_A\,\omega, \qquad \chi^\dagger = \chi, \qquad \chi^2 = 1_{16}, \qquad \chi\,\Gamma_A = -\Gamma_A\,\chi, \qquad \chi\,\Sigma_{AB} = \Sigma_{AB}\,\chi,$$
$$P_L + P_R = 1_{16}, \qquad P_L^2 = P_L, \quad P_R^2 = P_R, \quad P_LP_R = 0, \qquad \Gamma_AP_L = P_R\Gamma_A, \qquad P_L^\dagger\Gamma^0 = \Gamma^0P_R,$$
$$E_L^\dagger E_L = E_R^\dagger E_R = 1_8, \quad E_L^\dagger E_R = 0, \quad E_LE_L^\dagger = P_L, \quad E_RE_R^\dagger = P_R,$$
$$\chi E_L = -E_L, \quad \chi E_R = E_R, \quad W^\dagger W = 1_{16},$$
$$\mathrm{tr}\big(\Gamma_K^{-1}\Gamma_L\big) = 16\,\delta_{KL}, \qquad \Gamma_K^{-1} = \Gamma_K^\mathsf{T},$$
$$\textstyle\sum_K (\Gamma_K)_{ab}(\Gamma_K^{-1})_{cd} = 16\,\delta_{ad}\,\delta_{cb} \qquad \text{(Fierz completeness)}.$$

Blocks of the Dirac matrices between the chiral subspaces:

$$E_L^\dagger\beta E_R = E_R^\dagger\beta E_L = 1_8, \qquad E_L^\dagger\beta E_L = E_R^\dagger\beta E_R = 0,$$
$$E_L^\dagger\alpha_i E_L = -\sigma_i, \qquad E_R^\dagger\alpha_i E_R = \sigma_i, \qquad E_L^\dagger\alpha_i E_R = 0.$$

5. The field dirac16complex and its free dynamics

symbol	Mathematica	definition	meaning
ψ, ψ_a	`dirac16complex`, `ψ16[a]`	$\psi = (\psi_1, \ldots, \psi_{16})^{\mathsf{T}}$; `dirac16complex = Table[ψ16[a] @@ X, {a, 1, 16}]`	the quantized 16-component fermion field: complex Grassmann-odd operators
$\psi_\uparrow$, $\psi_\downarrow$	parts `[[1;;8]]` and `[[9;;16]]` of `dirac16complex`	$\psi = (\psi_\uparrow; \psi_\downarrow)$	upper and lower 8 components
$\bar\psi$	—	$\bar\psi = \psi^\dagger \Gamma^0$	Dirac adjoint
m	`mD`; `1` in the examples	$m > 0$	mass of the free field
$\Gamma^A \partial_A \psi$	`DiracOp17[ψ]`	$\sum_A \Gamma^A \partial_A \psi$	Dirac operator
$(i\Gamma^A \partial_A - m)\psi$	`DiracEq16[ψ, m]`	$i\Gamma^A \partial_A \psi - m\psi$	left side of the Dirac equation
h	`hD[ψ, m]`	$h = -i\alpha_i \partial_i + \beta m$	Dirac Hamiltonian: $i\partial_0 \psi = h\psi$
$h(\vec p)$	`hp[p, m]`	$h(\vec p) = p^i \alpha_i + m\beta$	Dirac Hamiltonian of a plane wave; real symmetric
$M(\vec p)$	`Mp`, `M0`, `M5`	$M(\vec p) = m 1_8 + p^i \tau_i^O$	real 8×8 block of $h(\vec p)$
$u_s(\vec p)$	`uS[s, M, E]`	$u_s = \frac{1}{\sqrt{E}}(E\, e_s;\ M^{\mathsf{T}} e_s)$	real positive-energy spinors
$v_s(\vec p)$	`vS[s, M, E]`	$v_s = \frac{1}{\sqrt{E}}(E\, e_s;\ -M e_s)$	real negative-energy spinors
$\not p$	`slash17[E, p]`	$\not p = \Gamma_A p^A = E\,\Gamma_0 + p^i \Gamma_i$	real 16×16 matrix
$U(\vec p)$	`Umat`	columns $u_s(\vec p)/\sqrt{2E}$ and $v_s(-\vec p)/\sqrt{2E}$, $s = 1, \ldots, 8$	real orthogonal matrix diagonalizing $h(\vec p)$
j^A	`jG` (Grassmann check)	$j^A = \bar\psi \Gamma^A \psi$	vector current; $j^0 = \psi^\dagger \psi$, $j^i = \psi^\dagger \alpha_i \psi$
$\mathcal{L}_0$	—	$\mathcal{L}_0 = \bar\psi (i\Gamma^A \partial_A - m)\psi$	free Lagrangian density
V	`Vbox`	—	volume of the periodic box of the mode expansion
$b_s(\vec p)$, $d_s(\vec p)$	`modeB`, `modeD` (coefficients)	operators of the mode expansion below	annihilate a particle, an antiparticle
H, Q	—	normal-ordered Hamiltonian and charge, below	energy and charge operators
ψ^c	—	$\psi^c = \omega\, \psi^*$	charge conjugate field
β_{b}	`β` (inside `Sβ`, `Λboost`)	real number	rapidity of a boost along x_1
$\mathcal{B}(\beta_{\mathrm{b}})$	`Sβ`, `Sβinv`	$\mathcal{B} = e^{\beta_{\mathrm{b}} \Sigma_{01}} = \cosh\frac{\beta_{\mathrm{b}}}{2}\, 1_{16} + \sinh\frac{\beta_{\mathrm{b}}}{2}\, \Gamma_0 \Gamma_1$	spinor boost; `Sβinv` is $\mathcal{B}^{-1}$
L	`Λboost`	$L_{BK} = \frac{1}{16}\mathrm{tr}\big(\eta_{BB} \Gamma_B \mathcal{B} \Gamma_K \mathcal{B}^{-1}\big)$	Lorentz matrix of the boost
p'	`pPrime`	$p' = Lp$	boosted 8-momentum
ϕ_L, ϕ_R	`ϕL`, `ϕR`	$\phi_L = E_L^\dagger \psi$, $\phi_R = E_R^\dagger \psi$	chiral (Weyl) 8-component fields
κ (field)	—	$\kappa = \phi_L^\dagger \phi_R$	complex scalar bilinear of the field

Equations that define and relate these symbols:

$$\big(i\,\Gamma^A \partial_A - m\big)\psi = 0, \qquad \big(i\Gamma^A \partial_A + m\big)\big(i\Gamma^A \partial_A - m\big) = -\big(\Box + m^2\big) 1_{16}, \qquad i\,\partial_0 \psi = h\,\psi,$$

$$h(\vec p) = \begin{pmatrix} 0 & M \\ M^{\mathsf{T}} & 0 \end{pmatrix}, \qquad M M^{\mathsf{T}} = M^{\mathsf{T}} M = E^2\, 1_8, \qquad M(-\vec p) = M(\vec p)^{\mathsf{T}}, \qquad h(\vec p)^2 = E^2\, 1_{16},$$

$$h(\vec p)\, u_s = E\, u_s, \qquad h(-\vec p)\, v_s = -E\, v_s, \qquad u_s^{\mathsf{T}} u_r = v_s^{\mathsf{T}} v_r = 2E\, \delta_{sr}, \qquad (\not p - m) u_s = 0, \qquad (\not p + m) v_s = 0,$$

$$\textstyle\sum_s u_s \bar u_s = \not p + m, \qquad \sum_s v_s \bar v_s = \not p - m, \qquad U^{\mathsf{T}} U = 1_{16}, \qquad U^{\mathsf{T}} h(\vec p)\, U = \mathrm{diag}(E\, 1_8, -E\, 1_8),$$

$$\partial_A j^A = 0, \qquad \psi^c \text{ solves the Dirac equation when } \psi \text{ does}, \qquad (\psi^c)^c = -\psi, \qquad L^{\mathsf{T}} \eta L = \eta.$$

Mode expansion in the box of volume V, anticommutators, Hamiltonian and charge:

$$\psi(x_0,\vec{x}) = \sum_{\vec{p}}\sum_{s=1}^{8} \frac{1}{\sqrt{2EV}}\Big[b_s(\vec{p})\,u_s(\vec{p})\,e^{-i(Ex_0-\vec{p}\cdot\vec{x})} + d_s^\dagger(\vec{p})\,v_s(\vec{p})\,e^{+i(Ex_0-\vec{p}\cdot\vec{x})}\Big],$$

$$\big\{b_s(\vec{p}), b_r^\dagger(\vec{q})\big\} = \big\{d_s(\vec{p}), d_r^\dagger(\vec{q})\big\} = \delta_{sr}\,\delta_{\vec{p}\vec{q}}, \qquad \big\{\psi_a(x_0,\vec{x}), \psi_b^\dagger(x_0,\vec{y})\big\} = \delta_{ab}\,\delta^7(\vec{x}-\vec{y}),$$

$$H = \int :\psi^\dagger h\,\psi:\, d^7x = \sum_{\vec{p},s} E\,\big(b_s^\dagger b_s + d_s^\dagger d_s\big), \qquad Q = \int :\psi^\dagger\psi:\, d^7x = \sum_{\vec{p},s}\big(b_s^\dagger b_s - d_s^\dagger d_s\big).$$

Chiral form of the field equations and the bilinears ($\psi = E_L\phi_L + E_R\phi_R$):

$$\big(i\,\partial_0 + \tau_i^O\,\partial_i\big)\,\phi_L = m\,\phi_R, \qquad \big(i\,\partial_0 - \tau_i^O\,\partial_i\big)\,\phi_R = m\,\phi_L, \qquad i\,\bar\sigma^A\partial_A\phi_L = m\,\phi_R, \quad i\,\sigma^A\partial_A\phi_R = m\,\phi_L,$$

$$\phi_L = \frac{\psi_\uparrow + i\,\psi_\downarrow}{\sqrt{2}}, \qquad \phi_R = \frac{i\,\psi_\uparrow + \psi_\downarrow}{\sqrt{2}}, \qquad \bar\psi\psi = \kappa + \kappa^*, \qquad \bar\psi\, i\chi\,\psi = i(\kappa - \kappa^*).$$

6. Grassmann variables

symbol	**Mathematica**	**definition**	**meaning**
θ_k	`θg[k]` = `grass[{k}]`	$\theta_k\theta_l = -\theta_l\theta_k$, hence $\theta_k^2 = 0$	Grassmann generators
$\theta_{k_1}\cdots\theta_{k_r}$	`grass[{k1, …, kr}]`	ordered monomial, $k_1 < \cdots < k_r$; `grass[{}]` $= 1$	basis of the Grassmann algebra
product	`gMul[x, y]`	bilinear extension of the product of monomials below	Grassmann product
$\partial^L/\partial\theta_k$	`gD[k, x]`	left derivative, below	moves θ_k to the front and removes it
$\bar\theta_k$	`grass[{3 + k}]`	$\bar\theta_k = \theta_{3+k}$	conjugate generators of the current-conservation check
U_1, U_2, U_3	`Ugr`	three plane-wave solutions, below	ingredients of a Grassmann-valued solution
ψ_G, $\bar\psi_G$	`ψG`, `ψbarG`	$\psi_G = \sum_k \theta_k U_k$, $\bar\psi_G = \sum_k \bar\theta_k U_k^\dagger\Gamma^0$	Grassmann-valued solution and its adjoint
ψ_a, $\bar\psi_a$ (32 generators)	`ψP`, `ψbarP`	$\psi_a = \theta_a$, $\bar\psi_a = \theta_{16+a}$	generators of the nonlinear-equation check
$\bar\psi M\psi$	`gBil[M]`	$\sum_{a,b}\bar\psi_a M_{ab}\psi_b$	Grassmann bilinear
$\mathcal{L}_{\rm int}$	`LintG`	$-\frac{g_S}{2}(\bar\psi\psi)^2 - \frac{g_P}{2}(\bar\psi i\chi\psi)^2 - \frac{g_V}{2}(\bar\psi\Gamma^A\psi)(\bar\psi\Gamma_A\psi) - \frac{g_A}{2}(\bar\psi\Gamma^A\chi\psi)(\bar\psi\Gamma_A\chi\psi)$	interaction Lagrangian (section 7)

Product and left derivative of monomials (K, K' ordered index sets; π the permutation that sorts the concatenation; the hat means omission):

$$\theta_K\,\theta_{K'} = \begin{cases} 0, & K\cap K' \neq \emptyset, \\ \operatorname{sign}(\pi)\,\theta_{\operatorname{sort}(K\cup K')}, & \text{otherwise}, \end{cases}$$

$$\frac{\partial^L}{\partial\theta_k}\big(\theta_{k_1}\cdots\theta_{k_r}\big) = (-1)^{j-1}\,\theta_{k_1}\cdots\widehat{\theta_{k_j}}\cdots\theta_{k_r} \quad \text{if } k = k_j \text{ (and 0 if } k\notin\{k_1,\ldots,k_r\}).$$

The three plane waves ($m = 1$):

$$U_1 = u_1(\vec{p}_1)\,e^{-i(5x_0-\vec{p}_1\cdot\vec{x})}, \qquad U_2 = v_3(\vec{p}_3)\,e^{+i(3x_0-\vec{p}_3\cdot\vec{x})}, \qquad U_3 = u_5(\vec{0})\,e^{-ix_0},$$

$$\vec{p}_1 = (2,2,4,0,0,0,0), \qquad \vec{p}_3 = (0,0,0,2,2,0,0).$$

7. The self-coupled field and the 16-component Kohn–Sham theory

symbol	Mathematica	definition	meaning
g_S, g_P, g_V, g_A	`gS`, `gP`, `gV`, `gA`	real constants of mass dimension -6	scalar, pseudoscalar, vector, axial-vector couplings; $g<0$ attractive
ρ_S	`dens16[Γ17[0]]`	$\rho_S = \bar\psi\psi$	scalar density
ρ_P	`dens16[M]`, $M=\Gamma^0 i\chi$	$\rho_P = \bar\psi\, i\chi\, \psi$	pseudoscalar density
j^A, j_A	`dens16[M]`, $M=\Gamma^0\Gamma^A$	$j^A = \bar\psi\Gamma^A\psi$, $j_A = \eta_{AA} j^A$	vector current
j_5^A, j_{5A}	`dens16[M]`, $M=\Gamma^0\Gamma^A\chi$	$j_5^A = \bar\psi\Gamma^A\chi\psi$, $j_{5A} = \eta_{AA} j_5^A$	axial-vector current
$\mathsf{s}, \mathsf{p}, \mathsf{v}_A, \mathsf{a}_A$	—	real functions of x	external sources coupled to ρ_S, ρ_P, j^A, j_5^A
n	—	$n = (\rho_S, \rho_P, j^A, j_5^A)$	the 18 densities $(1+1+8+8)$
w	—	$w = (\mathsf{s}, \mathsf{p}, \mathsf{v}_A, \mathsf{a}_A)$	the 18 sources
$w\cdot n$	—	$\mathsf{s}\rho_S + \mathsf{p}\rho_P + \mathsf{v}_A j^A + \mathsf{a}_A j_5^A$	source term
(g, O, O')	`vPairs`	$(g_S,1,1)$, $(g_P,i\chi,i\chi)$, (g_V,Γ^A,Γ_A), $(g_A,\Gamma^A\chi,\Gamma_A\chi)$, $A=0,\dots,7$	the 18 vertex triples $(1+1+8+8)$: coupling and the two matrices of a contact term $-\frac{g}{2}(\bar\psi O\psi)(\bar\psi O'\psi)$
$\mathcal{L}$	—	below	Lagrangian density of the self-coupled field
$\mathcal{H}$, $H[w]$, H_0, $H_{\rm int}$	—	below	Hamiltonian density and Hamiltonians
N	—	$Q = N$	number of particles (charge sector)
$F[n]$	—	$F[n] = \min_{\|\Psi\rangle\to n}\langle\Psi\|H_0+H_{\rm int}\|\Psi\rangle$	universal (Levy–Lieb) functional
$E[w]$	—	$E[w] = \min_n \left(F[n] + \int w\cdot n\, d^7x\right)$	ground-state energy
φ_k, ε_k	`orbV`, `orbcV`	$h_{\rm KS}\varphi_k = \varepsilon_k\varphi_k$, $\int \varphi_k^\dagger\varphi_l = \delta_{kl}$	Kohn–Sham orbitals and eigenvalues; `orbV`, `orbcV` are φ, φ^* as independent variables
$\mathcal{S}, \mathcal{P}$	`Sks`, `Pks`	below	scalar and pseudoscalar Kohn–Sham potentials
$\mathcal{V}_A, \mathcal{A}_A$	`VKS`, `AKS`	below; LOWER index	vector and axial Kohn–Sham potentials; `VKS[[A+1]]` is $\mathcal{V}_A$
$h_{\rm KS}$	`hKS[p]`	below	Kohn–Sham Hamiltonian; Hermitian for real potentials
T_s	—	$T_s = \sum_k \int \varphi_k^\dagger(-i\alpha_i\partial_i + \beta m)\varphi_k\, d^7x$	kinetic plus rest energy of the orbitals
E_H, e_H	`EH16`, `EHfun`	below	Hartree energy and its density
v_H	`vH16`	below	Hartree potential matrix
E_{xc}	—	$E_{xc} = F - T_s - E_H$	exchange-correlation energy
E_x, E_c	`Exfun[R]`	E_x below; $E_c = E_{xc} - E_x$	exact exchange energy; correlation energy
R	`Rorb`, `Rr`, `Runi`	$R = \sum_k \varphi_k\bar\varphi_k$ (16×16)	Kohn–Sham density matrix at a point
ϱ	—	$\varrho_{xy} = \langle c_y^\dagger c_x\rangle = \sum_k \varphi_{k,x}\varphi_{k,y}^*$, $R = \varrho\,\Gamma^0$	one-body density matrix
Σ_x	`Sigmax[R]`	below	exchange self-energy
Σ_H	—	$\Sigma_H = \mathcal{S} + i\chi\mathcal{P} + \Gamma^A(\mathcal{V}_A + \chi\mathcal{A}_A)$	Hartree self-energy
c_a, $c_a^\dagger$	`cdc[a, b]` $= c_a^\dagger c_b$	$\{c_a, c_b^\dagger\} = \delta_{ab}$	mode operators of the 16 components at one point
$B(M)$	`Bop[M]`	$B(M) = \sum_{a,b} M_{ab}\, c_a^\dagger c_b$	one-body operator, $\psi^\dagger M\psi$
$\|\Phi\rangle$	`Φ3`	Slater determinant of 3 orthonormal orbitals	state of the Wick-theorem check (`orbs3`; Fock space of dimension $\binom{16}{3} = 560$)

The Lagrangian and Hamiltonian densities, and the Hamiltonian with static sources:

$$\begin{aligned}
\mathcal{L} &= \bar\psi\big(i\Gamma^A\partial_A - m\big)\psi - \bar\psi\big(\mathsf{s} + i\chi\,\mathsf{p} + \Gamma^A(\mathsf{v}_A + \chi\,\mathsf{a}_A)\big)\psi \\
&\quad - \frac{g_S}{2}(\bar\psi\psi)^2 - \frac{g_P}{2}(\bar\psi\, i\chi\psi)^2 - \frac{g_V}{2}(\bar\psi\Gamma^A\psi)(\bar\psi\Gamma_A\psi) - \frac{g_A}{2}(\bar\psi\Gamma^A\chi\psi)(\bar\psi\Gamma_A\chi\psi), \\
\mathcal{H} &=: \psi^\dagger\big(-i\alpha_i\partial_i + \beta m\big)\psi : + : \bar\psi\big(\mathsf{s} + i\chi\,\mathsf{p} + \Gamma^A(\mathsf{v}_A + \chi\,\mathsf{a}_A)\big)\psi : \\
&\quad + \frac{g_S}{2} : \rho_S^2 : + \frac{g_P}{2} : \rho_P^2 : + \frac{g_V}{2} : j^A j_A : + \frac{g_A}{2} : j_5^A j_{5A} :, \\
H[w] &= H_0 + H_{\rm int} + \int w\cdot n\, d^7x, \qquad H_0 = \int : \psi^\dagger\big(-i\alpha_i\partial_i + \beta m\big)\psi :\, d^7x.
\end{aligned}$$

The Kohn–Sham Hamiltonian, potentials, densities and the Hartree quantities:

$$\begin{aligned}
h_{\rm KS} &= \alpha_i\big(-i\partial_i + \mathcal{V}_i\big) + \beta\big(m + \mathcal{S}\big) + \mathcal{V}_0 + \beta\, i\chi\,\mathcal{P} + \chi\,\mathcal{A}_0 + \alpha_i\chi\,\mathcal{A}_i, \\
\mathcal{S} &= \mathsf{s} + g_S\,\rho_S + \frac{\delta E_{xc}}{\delta\rho_S}, \qquad \mathcal{P} = \mathsf{p} + g_P\,\rho_P + \frac{\delta E_{xc}}{\delta\rho_P}, \\
\mathcal{V}_A &= \mathsf{v}_A + g_V\, j_A + \frac{\delta E_{xc}}{\delta j^A}, \qquad \mathcal{A}_A = \mathsf{a}_A + g_A\, j_{5A} + \frac{\delta E_{xc}}{\delta j_5^A}, \\
\rho_S &= \sum_k \bar\varphi_k\varphi_k, \qquad \rho_P = \sum_k \bar\varphi_k\, i\chi\,\varphi_k, \qquad j^A = \sum_k \bar\varphi_k\Gamma^A\varphi_k, \qquad j_5^A = \sum_k \bar\varphi_k\Gamma^A\chi\,\varphi_k, \\
E_H &= \int e_H\, d^7x, \qquad e_H = \frac{g_S}{2}\rho_S^2 + \frac{g_P}{2}\rho_P^2 + \frac{g_V}{2} j^A j_A + \frac{g_A}{2} j_5^A j_{5A}, \\
\frac{\partial e_H}{\partial\varphi_a^*} &= \big(v_H\varphi\big)_a \qquad \text{(one orbital } \varphi;\ \varphi,\varphi^* \text{ independent)}, \\
v_H &= \beta\, g_S\rho_S + \beta\, i\chi\, g_P\rho_P + \sum_A \Gamma^0\Gamma^A\, g_V j_A + \sum_A \Gamma^0\Gamma^A\chi\, g_A j_{5A}, \\
E &= \sum_{k=1}^N \varepsilon_k - \int\Big[(\mathcal{S}-\mathsf{s})\rho_S + (\mathcal{P}-\mathsf{p})\rho_P + (\mathcal{V}_A - \mathsf{v}_A)j^A + (\mathcal{A}_A - \mathsf{a}_A)j_5^A\Big]d^7x \\
&\quad + E_H + E_{xc}.
\end{aligned}$$

Exchange (Wick's theorem for contact couplings, $M_{1,2} = \Gamma^0 O_{1,2}$) and the Fierz rearrangement:

$$\begin{aligned}
\langle\Phi| : (\bar\psi O_1\psi)(\bar\psi O_2\psi) : |\Phi\rangle &= \mathrm{tr}(O_1R)\,\mathrm{tr}(O_2R) - \mathrm{tr}(O_1RO_2R), \\
:B(M_1)B(M_2): &= B(M_1)B(M_2) - B(M_1M_2), \\
E_x &= -\frac{g_S}{2}\mathrm{tr}(RR) - \frac{g_P}{2}\mathrm{tr}(i\chi R\, i\chi R) - \frac{g_V}{2}\mathrm{tr}(\Gamma^A R\Gamma_A R) - \frac{g_A}{2}\mathrm{tr}(\Gamma^A\chi R\Gamma_A\chi R), \\
\Sigma_x &= -g_S R - g_P(i\chi)R(i\chi) - g_V\Gamma_A R\Gamma^A - g_A\Gamma_A\chi R\Gamma^A\chi, \\
\delta E_x &= \mathrm{tr}\big(\Sigma_x\,\delta R\big), \qquad E_x = \tfrac12\,\mathrm{tr}\big(\Sigma_x R\big), \\
\mathrm{tr}\big(O_1RO_2R\big) &= \frac{1}{16}\sum_K \mathrm{tr}\big(\Gamma_K^{-1}R\big)\,\mathrm{tr}\big(O_1\Gamma_K O_2R\big).
\end{aligned}$$

8. The chiral Kohn–Sham theory

symbol	Mathematica	definition	meaning
φ_{kL}, φ_{kR}	`orbLV`, `orbRV` (and `orbLcV`, `orbRcV`)	$\varphi_{kL} = E_L^\dagger\varphi_k$, $\varphi_{kR} = E_R^\dagger\varphi_k$	chiral components of the Kohn–Sham orbitals
κ, κ^*	`κc`, `κcc`	$\kappa = \sum_k \varphi_{kL}^\dagger\varphi_{kR}$	complex scalar density
j_L^A	`jLc`	$j_L^A = \sum_k \varphi_{kL}^\dagger\bar\sigma^A\varphi_{kL}$	left-handed current
j_R^A	`jRc`	$j_R^A = \sum_k \varphi_{kR}^\dagger\sigma^A\varphi_{kR}$	right-handed current
ϖ	—	$\varpi = \mathsf{s} + i\,\mathsf{p}$	complex scalar source
$\mathcal{M}$	`Mks`	$\mathcal{M} = m + \mathcal{S} + i\,\mathcal{P}$	complex Kohn–Sham mass
$\mathcal{V}_A^L$, $\mathcal{V}_A^R$	`VLks`, `VRks`	$\mathcal{V}_A^L = \mathcal{V}_A - \mathcal{A}_A$, $\mathcal{V}_A^R = \mathcal{V}_A + \mathcal{A}_A$	chiral vector potentials (lower index)

symbol	Mathematica	definition	meaning
h_χ	`hChiKS[p]`	$h_\chi = W^\dagger h_{\rm KS} W$	chiral Kohn–Sham Hamiltonian, $8+8$ blocks
θ	—	real angle	chiral $(U(1)_A)$ rotation angle
$e^{i\theta\chi}$	`Uθ`	$e^{i\theta\chi} = \cos\theta\, 1_{16} - \sin\theta\,\omega$; `Uθ` has $\cos\theta = \frac{3}{5}$, $\sin\theta = \frac{4}{5}$	chiral rotation; a REAL matrix
$e^{2i\theta}$	`e2θ`	$(\frac{3}{5} + \frac{4}{5}i)^2$	phase of κ after the rotation `Uθ`

Equations of the chiral theory:

$$\rho_S = \kappa + \kappa^*, \qquad \rho_P = i\big(\kappa - \kappa^*\big), \qquad j^A = j_L^A + j_R^A, \qquad j_5^A = j_R^A - j_L^A,$$

$$\mathsf{s}\,\rho_S + \mathsf{p}\,\rho_P = \varpi\,\kappa + \varpi^*\kappa^*, \qquad \mathsf{v}_A j^A + \mathsf{a}_A j_5^A = (\mathsf{v}-\mathsf{a})_A j_L^A + (\mathsf{v}+\mathsf{a})_A j_R^A,$$

$$e_H = \frac{g_S - g_P}{2}\big(\kappa^2 + \kappa^{*2}\big) + (g_S + g_P)\,\kappa\kappa^* + \frac{g_V + g_A}{2}\big(j_L^A j_{LA} + j_R^A j_{RA}\big) + (g_V - g_A)\, j_L^A j_{RA},$$

$$h_\chi(\vec{p}) = \begin{pmatrix} \mathcal{V}_0^L - \sigma_i(p^i + \mathcal{V}_i^L) & \mathcal{M}\,1_8 \\ \mathcal{M}^*\,1_8 & \mathcal{V}_0^R + \sigma_i(p^i + \mathcal{V}_i^R) \end{pmatrix},$$

$$\mathcal{M} - m = \varpi + \frac{\partial(e_H + e_{xc})}{\partial\kappa},$$

$$\mathcal{V}_A^L = (\mathsf{v}-\mathsf{a})_A + \frac{\partial(e_H + e_{xc})}{\partial j_L^A}, \qquad \mathcal{V}_A^R = (\mathsf{v}+\mathsf{a})_A + \frac{\partial(e_H + e_{xc})}{\partial j_R^A},$$

$$e^{i\theta\chi}: \quad \varphi_L \to e^{-i\theta}\varphi_L, \quad \varphi_R \to e^{i\theta}\varphi_R, \quad \kappa \to e^{2i\theta}\kappa, \quad j_L,\ j_R \text{ invariant.}$$

9. Uniform matter and the Dirac sea

symbol	Mathematica	definition	meaning
k_F	`kF`; `kF0` at saturation	—	Fermi momentum
m^*	`gap7[g, kF]`; `ms0`, `msHF`	$m^* = m + \mathcal{S}$, solution of the gap equation below	effective (Kohn–Sham) mass
E^*	—	$E^* = \sqrt{\vec{p}^{\,2} + m^{*2}}$	energy of a Kohn–Sham plane wave
$\lvert S^6\rvert$	—	$\lvert S^6\rvert = 2\pi^{7/2}/\Gamma_{\rm E}(\frac{7}{2}) = \frac{16\pi^3}{15}$	area of the unit sphere $S^6 \subset \mathbb{R}^7$ ($\Gamma_{\rm E}$ is Euler's Gamma function)
c_7	`c7`	$c_7 = \frac{8\,\lvert S^6\rvert}{(2\pi)^7} = \frac{1}{15\pi^4}$	phase-space factor for 8 spin states in 7 dimensions
ρ	`rho7[kF]`; `rho0`	$\rho = j^0 = \frac{k_F^7}{105\pi^4}$	particle density
ρ_S	`c7 iS7[ms, kF]`; `rhoS0`	integral below	scalar density of uniform matter
$\varepsilon_{\rm kin}$	`c7 iE7[ms, kF]`	integral below	kinetic plus rest energy density
ε	`eps7[g, gv, kF]`	below	energy density
E/A	`eps7[…]/rho7[kF]`	$E/A = \varepsilon/\rho$	energy per particle
ε_F	`fermi7[g, gv, kF]`	$\varepsilon_F = \mathcal{V}_0 + \sqrt{k_F^2 + m^{*2}}$	Fermi energy
μ	`dEdrho7[g, gv, kF]`	$\mu = d\varepsilon/d\rho$	chemical potential (numerical derivative with step 10^{-12} in the notebook)
$\mathcal{V}_0$	`V00` at saturation	$\mathcal{V}_0 = g_V\rho$ (Hartree)	vector potential of uniform matter
G_S, G_V	`GSeff`, `GVeff`; `GS1`, `GV1`	below	Fierz-transformed couplings of uniform Hartree–Fock
S_x, V_x	—	$S_x = \partial E_x/\partial\rho_S$, $V_x = \partial E_x/\partial\rho$	uniform exchange potentials, $\Sigma_x = S_x 1_{16} + V_x\Gamma_0$

symbol	Mathematica	definition	meaning
Λ	`Λc`	$\Lambda = 1$ in the example	momentum cutoff of the Dirac sea, $\lvert\vec{p}\rvert < \Lambda$
$J(M)$, $K(M)$	`Jsea[M]`, `Ksea[M]`	integrals below	sea integrals
g	`gNJL`	$g = g_S = g_P$; $g = \frac{6}{5} g_c$ in the example	chirally symmetric coupling
g_c	`gcrit`	$g_c = -\frac{1}{8J(0)} = -\frac{90\pi^4}{\Lambda^6}$	critical coupling
$h_\chi(\vec{p})$	`hChiSea`	$\begin{pmatrix} -\sigma\cdot\vec{p} & \mathcal{M} 1_8 \\ \mathcal{M}^* 1_8 & \sigma\cdot\vec{p} \end{pmatrix}$	chiral Hamiltonian of the vacuum
$\Pi_-(\vec{p})$	`PiMinus`	$\Pi_- = \frac{1}{2}\big(1_{16} - h_\chi/E\big)$, $E = \sqrt{\vec{p}^2 + \lvert\mathcal{M}\rvert^2}$	projector on the filled Dirac sea
$\mathcal{E}(\mathcal{M})$	`Esea[M, m, g]`	below	vacuum energy density
ϕ	`phi` (loop variable)	$\mathcal{M} = \lvert\mathcal{M}\rvert e^{i\phi}$	phase of the chiral condensate (Goldstone direction)

Equations of uniform matter (no sea; units $m = 1$ in the examples):

$$8\int \frac{d^7k}{(2\pi)^7}\, f(|\vec{k}|) = c_7 \int_0^\infty k^6 f(k)\, dk,$$

$$\rho_S = c_7 \int_0^{k_F} \frac{k^6\, m^*}{\sqrt{k^2 + m^{*2}}}\, dk, \qquad \varepsilon_{\rm kin} = c_7 \int_0^{k_F} k^6 \sqrt{k^2 + m^{*2}}\, dk,$$

$$m^* = m + g_S\, \rho_S(m^*), \qquad \varepsilon = \varepsilon_{\rm kin}(m^*) - \frac{(m^* - m)^2}{2 g_S} + \frac{g_V}{2}\rho^2, \qquad \mu = \frac{d\varepsilon}{d\rho} = \varepsilon_F,$$

$$R = \frac{\rho_S\, 1_{16} + \rho\, \Gamma_0}{16} \quad \text{(uniform matter)},$$

$$G_S = g_S + \frac{-g_S + g_P - 8 g_V + 8 g_A}{16}, \qquad G_V = g_V + \frac{-g_S - g_P + 6 g_V + 6 g_A}{16}.$$

Equations of the Dirac sea (chiral vacuum, $g_P = g_S = g$):

$$J(M) = \frac{1}{120\pi^4}\int_0^\Lambda \frac{p^6\, dp}{\sqrt{p^2 + M^2}}, \qquad K(M) = \frac{1}{120\pi^4}\int_0^\Lambda p^6 \sqrt{p^2 + M^2}\, dp,$$

$$J(0) = \frac{\Lambda^6}{720\pi^4}, \qquad K(0) = \frac{\Lambda^8}{960\pi^4}, \qquad \kappa = -4\,\mathcal{M}^* J(|\mathcal{M}|), \qquad \mathcal{M} = m - 8\, g\, J(|\mathcal{M}|)\, \mathcal{M},$$

$$\mathcal{E}(\mathcal{M}) = -8K(|\mathcal{M}|) + 8J(|\mathcal{M}|)\,\mathrm{Re}\big[(\mathcal{M} - m)\mathcal{M}^*\big] + 32\, g\, |\mathcal{M}|^2 J(|\mathcal{M}|)^2.$$

Numerical parameters of the examples: `gS0` $= g_S = -10^5$, `gV0` $= g_V = 10^4$ ($g_P = g_A = 0$, $m = 1$); `wp` $= 50$ (digits of the inner root finding); `mq0` $= m = \frac{1}{100}$ (small explicit mass of the sea example); `M0NJL` and `MmNJL` are $|\mathcal{M}|$ for $m = 0$ and for $m = \frac{1}{100}$; `pS` $= \vec{p} = (\frac{3}{10}, -\frac{1}{5}, \frac{1}{10}, \frac{1}{4}, 0, -\frac{3}{20}, \frac{1}{20})$ and `Mz7` $= \mathcal{M} = |\mathcal{M}|e^{7i/10}$ are the momentum and the complex mass at which $h_\chi(\vec{p})$ is diagonalized.

10. Symbols that look alike

symbols	how to tell them apart
$\{7, 1, 0\}$, η	$\{7, 1, 0\}$ is the xAct signature list of section 12 of 7+1.nb: seven space-like dimensions, one time-like dimension, none degenerate; $\eta = \mathrm{diag}(+1, -1, \ldots, -1)$ is simply the squares of the octonion basis, $\eta_{AB} = \mathrm{Re}(e_A e_B)$.
β, $\beta_{\rm b}$, `βD`, `β`	$\beta = \Gamma^0$ is the Dirac matrix (`βD`); $\beta_{\rm b}$ is the rapidity of a boost (the plain symbol `β` in `Sβ` and `Λboost`).
L, $\mathcal{L}$, Λ, `Λboost`, `Λc`	L is the 8×8 Lorentz matrix of a boost (Mathematica `Λboost`); $\mathcal{L}$ is a Lagrangian density; Λ is the momentum cutoff (`Λc`). The subscript L means left-handed.
$\mathcal{B}(\beta_{\rm b})$, $B(M)$	$\mathcal{B}$ is the spinor boost; $B(M)$ the one-body operator $\sum M_{ab} c_a^\dagger c_b$; B is also a spacetime index.

symbols	how to tell them apart
$\mathcal{S}$, s, s, S_x, S^6	$\mathcal{S}$ scalar Kohn–Sham potential; s scalar source; s spin label; S_x uniform scalar exchange potential; S^6 unit sphere.
p, p, $\vec{p}$	p pseudoscalar source; $p = (E, \vec{p})$ 8-momentum; $\vec{p}$ momentum.
a_A, $\mathcal{A}_A$, a	axial source; axial Kohn–Sham potential; spinor index.
v_A, $\mathcal{V}_A$, V, V_x	vector source; vector Kohn–Sham potential; box volume; uniform vector exchange potential.
μ, ϖ	μ is the chemical potential $d\varepsilon/d\rho$; $\varpi = \mathsf{s} + i\mathsf{p}$ is the complex scalar source.
$M(\vec{p})$, $\mathcal{M}$, M in $J(M)$, $M_{1,2}$	real 8×8 block of $h(\vec{p})$; complex Kohn–Sham mass; argument of the sea integrals; matrices $\Gamma^0 O_{1,2}$ of Wick's theorem.
K in Γ_K, $K(M)$, index K	ordered index set of a basis element; sea integral; spacetime index.
$J(M)$, j^A, j_5^A, j_L^A, j_R^A	sea integral; vector, axial, left and right currents.
κ (field), κ (Kohn–Sham)	the same bilinear $\phi_L^\dagger \phi_R$, evaluated for the field or summed over the Kohn–Sham orbitals.
ϕ_L, ϕ_R, φ_k, $\varphi_{kL}, \varphi_{kR}$, ϕ	chiral fields; Kohn–Sham orbitals; their chiral components; phase of $\mathcal{M}$.
θ_k, θ	Grassmann generators; chiral rotation angle.
σ, σ_i, σ^A, $\bar{\sigma}^A$	8×8 matrix $\tau_1\tau_2\tau_3$; Pauli matrices $i\tau_i^O$; chiral 8×8 matrices.
Σ_{AB}, Σ_x, Σ_H	Lorentz generators; exchange self-energy; Hartree self-energy.
Γ_A, Γ_{E}	Dirac matrices; Euler's Gamma function (only in $\lvert S^6 \rvert$).
E, E^*, E_L, E_R, E_H, E_x, E_{xc}, $E[w]$, E/A, $\mathcal{E}$	free energy $\sqrt{m^2 + \vec{p}^2}$; Kohn–Sham plane-wave energy; chiral isometries; Hartree, exchange, exchange-correlation energies; ground-state energy; energy per particle; vacuum energy density.
ε, ε_k, ε_F, $\varepsilon_{\mathrm{kin}}$	energy density; Kohn–Sham eigenvalues; Fermi energy; kinetic energy density.
e_s, e_H, e_i	unit column of length 8; Hartree energy density; octonion units.
ρ, ρ_S, ρ_P, ϱ	particle density j^0; scalar and pseudoscalar densities; one-body density matrix.
R, ϱ	$R = \varrho\,\Gamma^0$, both 16×16.
h, $h(\vec{p})$, h_{KS}, h_χ	Dirac Hamiltonian; its plane-wave form; Kohn–Sham Hamiltonian; its chiral form.
H, $\mathcal{H}$, $H[w]$, $\mathbb{H}$	Hamiltonian; Hamiltonian density; Hamiltonian with sources; quaternions.
Q, $Q(v)$	charge; quadratic form $v \cdot v$ of the Clifford algebra.
T, T_A, T_s	invertible matrix of the proof in section 4; Dirac matrices of 4+4.nb; Kohn–Sham kinetic energy.
Y, Z, R_A, R	matrices of the two proofs in section 4; the Kohn–Sham density matrix R.
Γ_A and T_A	$\Gamma_0 = T_0$, $\Gamma_i = T_0 T_i$ for $i = 1, 2, 3$, $\Gamma_k = T_0 T_k T_4 T_5 T_6 T_7$ for $k = 4, \ldots, 7$.
c_7, c_a	phase-space factor; mode operators.
w, W	collection of sources; chiral basis change.
N, n	number of particles; collection of densities.
δ_{ij}, δ^7, $\delta E/\delta\rho$	Kronecker delta; Dirac delta; functional derivative.
g, g_S, g_P, g_V, g_A, g_c, G_S, G_V	common coupling $g_S = g_P$ of the sea example; the four couplings; critical coupling; Fierz-transformed couplings.

11. Every Mathematica name of sections 7–10 of 7+1.nb

The names are listed in the order of the notebook sections in which they are defined. Names defined in sections 0–6 of the notebook and used here: `X`, `η17`, `η17inv`, `ID8`, `ID16`, `Zero8`, `Zero16`, `zero8`, `τ[A]`, `τO[i]`, `σ`, `Γ17[A]` ($A = 0, \ldots, 8$; `Γ17[8]` $= \omega$), `Γ17up[A]`, `products256`, `χ17`, `PL17`, `PR17`, `Σ17[A, B]`, `Sβ`, `Sβinv`, `Λboost`, `DiracOp17`, `waveOp17`, `check`, `checks`. The symbols `mD` (m), `p1`, …, `p7` (p^i), `gS`, `gP`, `gV`, `gA`, `Sks`, `Pks`, `Vks[A]`, `Aks[A]`, `Vbox`, `β`, `ρSsym`, `ρsym`, `V0s`, `mst`, `εs` are left without values (symbolic).

11.1 Section 7: the Dirac equation of dirac16complex

name	Wolfram Language definition (abbreviated)	meaning
βD, αD[i], ω17	βD = Γ17up[0]; αD[i] = Γ17up[0].Γ17up[i]; ω17 = Γ17[8]	β, α_i, ω
dirac16complex	Table[ψ16[a] @@ X, {a, 1, 16}]	ψ: 16 arbitrary functions of $x_0, \ldots, x_7$
DiracEq16[ψ, m]	I DiracOp17[ψ] - m ψ	$(i\Gamma^A\partial_A - m)\psi$
hD[ψ, m]	Sum[-I αD[i].D[ψ, X[[i+1]]], {i,1,7}] + m βD.ψ	$h\psi$
pv7, asm7	{p1, …, p7}; mD > 0 && p1, …, p7 real	$\vec{p}$; assumptions for Simplify
Ep, Mp	Sqrt[mD^2 + pv7.pv7]; mD ID8 + Sum[pv7[[i]] τO[i], {i,1,7}]	E, $M(\vec{p})$
hp[q, m]	Sum[q[[i]] αD[i], {i,1,7}] + m βD	$h(\vec{q})$
uS[s, M, E], vS[s, M, E]	Join[E e_s, Transpose[M].e_s]/Sqrt[E]; Join[E e_s, -M.e_s]/Sqrt[E]	u_s, v_s
slash17[E, q]	E Γ17[0] + Sum[q[[i]] Γ17[i], {i,1,7}]	$\not{p}$
p0v, M0, u0Matrix	{2, 2, 4, 0, 0, 0, 0}; ID8 + p0v.τO; Transpose[Table[uS[s, M0, 5], {s,1,8}]]	example momentum ($m = 1$, $E = 5$), M, the matrix $(u_1, \ldots, u_8)$
USmat, VSmat, VSmatMinus	columns $u_s(\vec{p})$, $v_s(\vec{p})$, $v_s(-\vec{p})$ (with $M(-\vec{p}) = M^{\mathsf{T}}$)	16×8 matrices of spinors
Umat	Join[USmat, VSmatMinus, 2]/Sqrt[2 Ep]	$U(\vec{p})$
grass[{k1, …}]	ordered monomial (inert head)	$\theta_{k_1} \cdots \theta_{k_r}$
gTerms, gSplit	list of the terms of an expanded sum; {coefficient, monomial} of a term	helpers of gMul, gD
gProd	0 if two monomials share a generator, else Signature[...] grass[Sort[...]]	product of monomials
gMul[x, y]	bilinear extension of gProd	Grassmann product
gD[k, x]	left derivative	$\partial^L x/\partial\theta_k$
θg[k]	grass[{k}]	θ_k
conjR[e]	e /. Complex[r, s] :> Complex[r, -s]	complex conjugate of an expression whose symbols are real
mom5, E5, M5	momenta $\pm(2,2,4,0,0,0,0,0)$, $\pm(0,0,0,2,2,0,0,0)$, $\vec{0}$; energies $5,5,3,3,1$; $M(\vec{p}_n)$	five modes ($m = 1$)
Ugr	{uS[1, M5[[1]], 5] e^{-i(5 x0 - p1.x)}, vS[3, M5[[3]], 3] e^{i(3 x0 - p3.x)}, uS[5, M5[[5]], 1] e^{-i x0}}	U_1, U_2, U_3
ψG, ψbarG, jG	$\sum_k \theta_k U_k$; $\sum_k \theta_{3+k} U_k^\dagger \Gamma^0$; $\bar{\psi}_G \Gamma^A \psi_G$ with gMul	Grassmann-valued solution, adjoint, current
xv7, yv7	Rest[X]; Array[y7, 7]	$\vec{x}$, $\vec{y}$
modeB[n, s, x], modeD[n, s, x]	uS[…] Exp[-I E x0 + I p.x]/Sqrt[2 E Vbox]; vS[…] Exp[I E x0 - I p.x]/Sqrt[2 E Vbox]	coefficients of $b_s(\vec{p}_n)$ and $d_s^\dagger(\vec{p}_n)$ in ψ
carMatrix	$\sum_{n,s}$ [modeB ⊗ conjR[modeB] + modeD ⊗ conjR[modeD]]	$\{\psi_a(\vec{x}), \psi_b^\dagger(\vec{y})\}$ from the mode algebra
ψreF, ψimF	16 arbitrary real functions each	real and imaginary parts of ψ (charge-conjugation check)
Sb0, Λb0, pPrime	Sβ, Λboost at $\beta_{\mathrm{b}} = 2\log 2$; Λb0.{5, 2, 2, 4, 0, 0, 0, 0}	$\mathcal{B}$, L, $p' = (\frac{55}{8}, -\frac{41}{8}, 2, 4, 0, 0, 0, 0)$
EL, ER, Wch	Join[ID8, -I ID8]/Sqrt[2]; Join[-I ID8, ID8]/Sqrt[2]; Join[EL, ER, 2]	E_L, E_R, W
ϕL, ϕR	ConjugateTranspose[EL].dirac16complex; same with ER	ϕ_L, ϕ_R
σW[i], σWp[q]	I τO[i]; Sum[q[[i]] σW[i], {i,1,7}]	σ_i, $\sigma \cdot \vec{q}$
tauD[f]	Sum[τO[i].D[f, X[[i+1]]], {i,1,7}]	$\tau_i^O \partial_i f$
f8	Table[f8c[a] @@ X, {a, 1, 8}]	8 arbitrary functions (Klein–Gordon check of the chiral equations)

11.2 Section 8: the 16-component Kohn–Sham theory

name	Wolfram Language definition (abbreviated)	meaning
vPairs	{{gS, ID16, ID16}, {gP, I χ17, I χ17}} + {gV, Γ17up[A], Γ17[A]} + {gA, Γ17up[A].χ17, Γ17[A].χ17}	the 18 vertex triples (g, O, O')
ψP, ψbarP	grass[{a}]; grass[{16 + a}]	32 Grassmann generators
gBil[M]	$\sum_{a,b} M_{ab}$ gMul[ψbarP[[a]], ψP[[b]]]	$\bar{\psi} M \psi$

name	Wolfram Language definition (abbreviated)	meaning
LintG	Sum[-g/2 gMul[gBil[O], gBil[Oc]], {g, O, Oc} in vPairs]	$\mathcal{L}_{\text{int}}$
VKS, AKS	Array[Vks, 8, 0]; Array[Aks, 8, 0]	$\mathcal{V}_A$, $\mathcal{A}_A$ (lower index)
hKS[q]	Sum[αD[i] (q[[i]] + VKS[[i+1]])] + βD (mD + Sks) + VKS[[1]] ID16 + βD.(I χ17) Pks + χ17 AKS[[1]] + Sum[αD[i].χ17 AKS[[i+1]]]	$h_{\text{KS}}(\vec{q})$
orbV, orbcV	Array[orb, 16]; Array[orbc, 16]	φ and φ^* (independent variables)
dens16[M]	orbcV.M.orbV	$\varphi^\dagger M \varphi$
EHfun[d], EH16	$\frac{g_S}{2} d[\Gamma^0]^2 + \frac{g_P}{2} d[\Gamma^0 i\chi]^2 + \frac{g_V}{2} \sum_A \eta_{AA} d[\Gamma^0\Gamma^A]^2 + \frac{g_A}{2} \sum_A \eta_{AA} d[\Gamma^0\Gamma^A\chi]^2$; EHfun[dens16]	e_H for a density function d; for one orbital
vH16	$\beta g_S \rho_S + \beta i\chi g_P \rho_P + \sum_A \eta_{AA}\Gamma^0\Gamma^A g_V j^A + \sum_A \eta_{AA}\Gamma^0\Gamma^A\chi g_A j_5^A$	v_H
basis3, idx3	Subsets[Range[16], {3}]; position of each subset	the 560 three-particle basis states
cdc[a, b]	sparse 560×560 matrix with signs from the anticommutation	$c_a^\dagger c_b$ on three-particle states
Bop[M]	Sum[M[[a, b]] cdc[a, b]]	$B(M)$
gsOrth	Gram–Schmidt orthonormalization of a list of vectors	helper
orbs3, Φ3	3 random orthonormal orbitals (SeedRandom[71], 40 digits); their Slater determinant	state of the Wick check
Rorb	Transpose[orbs3].Conjugate[orbs3].Γ17[0]	$R = \sum_k \varphi_k \bar{\varphi}_k$
wickLHS[o1, o2]	$\langle\Phi\|B(M_1)B(M_2)\|\Phi\rangle - \langle\Phi\|B(M_1M_2)\|\Phi\rangle$, $M_i = \Gamma^0 o_i$	left side of Wick's theorem
Rr, Dr	random complex integer 16×16 matrices (entries $-3,\ldots,3$)	test matrices for R and δR
Exfun[R], Sigmax[R]	Sum[-g/2 Tr[O.R.Oc.R]]; Sum[-g Oc.R.O] over vPairs	E_x, Σ_x
F256	Flatten /@ products256	the 256 basis matrices as vectors (Fierz completeness)
Runi, ExUni	(ρSsym ID16 + ρsym Γ17[0])/16; the formula of E_x for uniform matter	R and E_x of uniform matter
GSeff[gs, gp, gv, ga], GVeff[…]	gs + (-gs + gp - 8 gv + 8 ga)/16; gv + (-gs - gp + 6 gv + 6 ga)/16	G_S, G_V
c7, rho7[kF]	1/(15 Pi^4); kF^7/(105 Pi^4)	c_7, ρ
iS7[ms, kF], iE7[ms, kF]	Integrate[k^6 ms/Sqrt[k^2 + ms^2], {k, 0, kF}]; Integrate[k^6 Sqrt[k^2 + ms^2], {k, 0, kF}] (closed forms)	ρ_S/c_7, $\varepsilon_{\text{kin}}/c_7$
gS0, gV0, wp	-10^5, 10^4, 50	couplings of the example; working precision
gap7[g, kF]	root of ms - 1 - g c7 iS7[ms, kF] (Brent, 50 digits)	m^*
eps7[g, gv, kF]	c7 iE7[m*, kF] - (m* - 1)^2/(2 g) + gv/2 rho7[kF]^2	ε
fermi7[g, gv, kF]	gv rho7[kF] + Sqrt[kF^2 + m*^2]	ε_F
dEdrho7[g, gv, k0]	symmetric difference quotient with $h = 10^{-12}$	$d\varepsilon/d\rho$
kF0	root of eps7/rho7 - fermi7 in $[0.7, 0.8]$ (40 digits)	saturation k_F
ms0, rho0, rhoS0, V00, eps0	gap7[gS0, kF0], rho7[kF0], c7 iS7[ms0, kF0], gV0 rho0, eps7[gS0, gV0, kF0]	saturation values
hartreeSaturation, hartreeKF1	rules with the printed numbers	tables of section 8.7 of the notebook
GS1, GV1, msHF, rhoHF, rhoSHF	GSeff[gS0, 0, gV0, 0], GVeff[…]; m^*, ρ, ρ_S at $k_F = \frac{3}{4}$	Hartree–Fock example
hartreeFock34, eAcurve[g, gv]	rules with the printed numbers; list of $(k_F, E/A - 1)$	Hartree–Fock table; data of the plot

11.3 Section 9: the chiral Kohn–Sham theory

name	Wolfram Language definition (abbreviated)	meaning
orbLV, orbLcV, orbRV, orbRcV	Array[orbL, 8], Array[orbLc, 8], Array[orbR, 8], Array[orbRc, 8]	φ_L, φ_L^*, φ_R, φ_R^*
psiChi, psiChiDag	EL.orbLV + ER.orbRV; orbLcV.EL^† + orbRcV.ER^†	$\varphi = E_L\varphi_L + E_R\varphi_R$ and $\varphi^\dagger$
densChi[M]	psiChiDag.M.psiChi	$\varphi^\dagger M \varphi$ in chiral variables
κc, κcc	orbLcV.orbRV; orbRcV.orbLV	κ, κ^*
σUp, σbarUp	{ID8, σW[1], …, σW[7]}; {ID8, -σW[1], …, -σW[7]}	σ^A, $\bar{\sigma}^A$
jLc, jRc	orbLcV.σbarUp[[A+1]].orbLV; orbRcV.σUp[[A+1]].orbRV	j_L^A, j_R^A
EHchiAbs[k, kc, jl, jr], EHchi	$\frac{g_S}{2}(k + k_c)^2 - \frac{g_P}{2}(k - k_c)^2 + \frac{g_V + g_A}{2}(j_l \cdot j_l + j_r \cdot j_r) + (g_V - g_A)\, j_l \cdot j_r$; at κc, κcc, jLc, jRc	e_H in chiral variables

name	Wolfram Language definition (abbreviated)	meaning
VLks, VRks, Mks	VKS - AKS, VKS + AKS, mD + Sks + I Pks	$\mathcal{V}_A^L$, $\mathcal{V}_A^R$, $\mathcal{M}$
hChiKS[q]	{{VL0 ID8 - Sum[σW[i] (q[[i]] + VLi)], Mks ID8}, {(mD + Sks - I Pks) ID8, VR0 ID8 + Sum[σW[i] (q[[i]] + VRi)]}}	$h_\chi(\vec{q})$
VLc, VRc	$(g_V+g_A)j_{LA}+(g_V-g_A)j_{RA}$; $(g_V+g_A)j_{RA}+(g_V-g_A)j_{LA}$	Hartree $\mathcal{V}_A^L$, $\mathcal{V}_A^R$
dEHdk, dEHdkc	$g_S(\kappa+\kappa^*)-g_P(\kappa-\kappa^*)$; $g_S(\kappa+\kappa^*)+g_P(\kappa-\kappa^*)$	$\partial e_H/\partial\kappa$, $\partial e_H/\partial\kappa^*$
U0, e2θ	3/5 ID16 + I 4/5 χ17; (3/5 + 4/5 I)^2	$e^{i\theta\chi}$ with $\cos\theta=\frac{3}{5}$; $e^{2i\theta}$
EHrot	e_H of the rotated orbital $e^{i\theta\chi}\varphi$	chiral-rotation check
κ16, κ16c	dens16[EL.ER^†]; dens16[ER.EL^†]	κ, κ^* from a 16-component orbital
kappaUni, jLUni, jRUni	traces with Runi	κ, j_L^A, j_R^A of uniform matter
hChiUni	{{V0s ID8 - σWp[pv7], mst ID8}, {mst ID8, V0s ID8 + σWp[pv7]}}	h_χ of uniform matter ($\mathcal{V}_0$, m^* symbolic)
chiralUniform	rules with the printed numbers	table of section 9.5 of the notebook
Λc, Jsea[M], Ksea[M]	1; closed forms of $J(M)$, $K(M)$	cutoff and sea integrals
J0sea, K0sea, gcrit	$J(0)$, $K(0)$; -1/(8 J0sea)	$J(0)$, $K(0)$, g_c
Esea[Mz, mq, g], E0sea	-8 Ksea[|Mz|] + 8 Jsea[|Mz|] Re[(Mz - mq) Mz^*] + 32 g |Mz|^2 Jsea[|Mz|]^2; -8 K0sea	$\mathcal{E}(\mathcal{M})$; $\mathcal{E}(0)$ for $m=0$
gNJL, gapNJL[M, mq]	6/5 gcrit; M - mq + 8 gNJL Jsea[M] M	g; the gap equation for real $\mathcal{M}>0$
M0NJL, mq0, MmNJL	root of gapNJL[M, 0]/M; 1/100; root of gapNJL[M, mq0]	$\lvert\mathcal{M}\rvert$ for $m=0$; m; $\mathcal{M}$ for $m=\frac{1}{100}$
njlVacuum	rules with the printed numbers	table of section 9.6 of the notebook
pS, Mz7, Esp	$(\frac{3}{10},-\frac{1}{5},\frac{1}{10},\frac{1}{4},0,-\frac{3}{20},\frac{1}{20})$; M0NJL Exp[7 I/10]; $\sqrt{\vec{p}^2+\lvert\mathcal{M}\rvert^2}$	test momentum, complex mass, energy
hChiSea, PiMinus, h0Chi	$h_\chi(\vec{p})$; $\frac{1}{2}(1_{16}-h_\chi/E)$; h_χ with $\mathcal{M}=0$	sea Hamiltonian, projector, massless Hamiltonian

Section 10 of the notebook defines no new names; it prints Length[checks], failedChecks, allChecksPassed and a table of all checks.